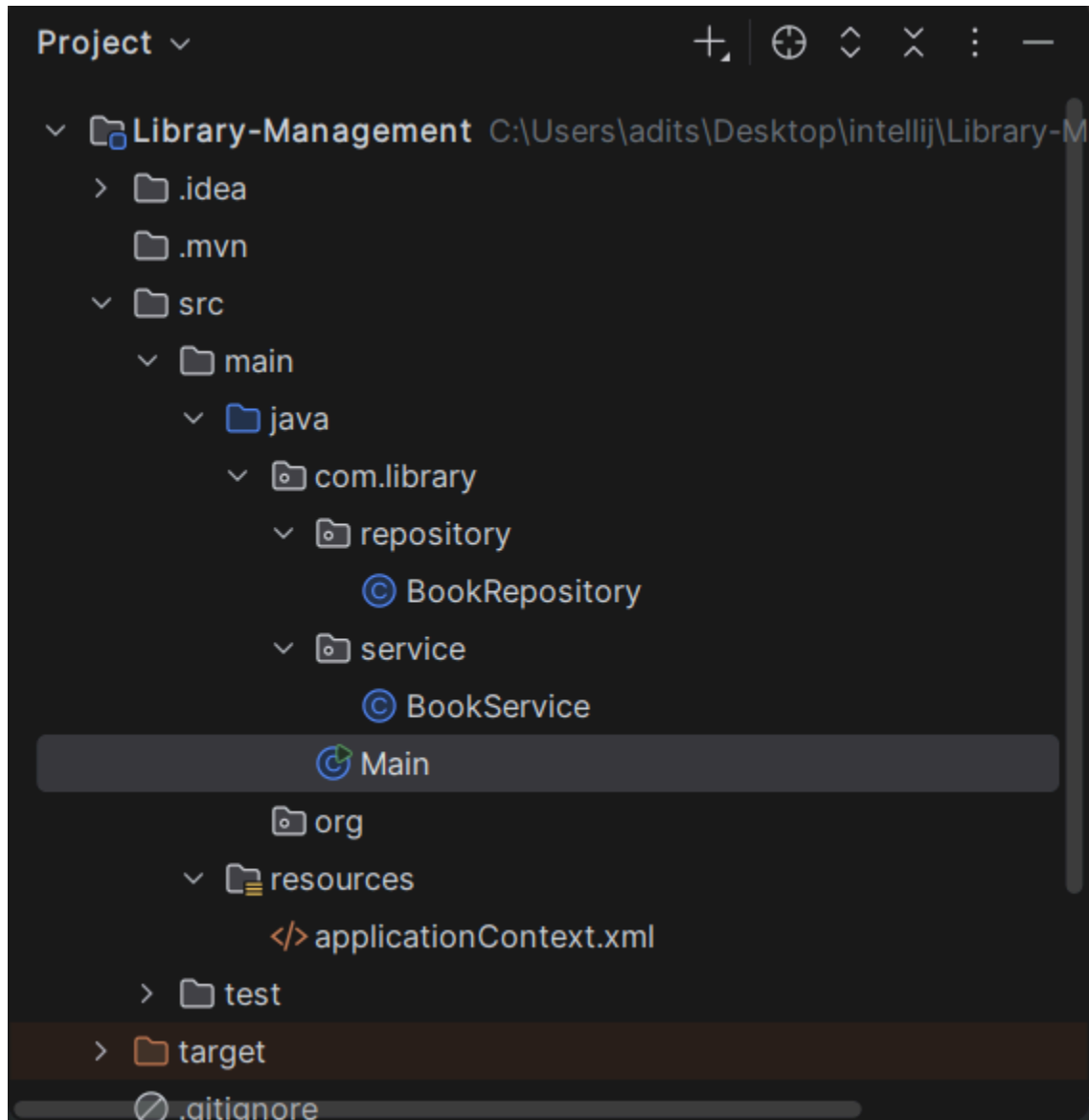


Exercise 5: Configuring the Spring IoC Container

File Structure:



Code:

BookRepository.java:

```
package com.library.repository;  
  
public class BookRepository {
```

```

    public void showRepository() {
        System.out.println("Book Repository is working.");
    }
}

```

BoorSerice.java

```

package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    // Setter Injection
    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void showService() {
        System.out.println("Book Service is working.");
        bookRepository.showRepository();
    }

}

```

Pom.xml:

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>org.example</groupId>
    <artifactId>Library-Management</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>25</maven.compiler.source>
        <maven.compiler.target>25</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>

```

```

<dependencies>

    <dependency>
        <groupId>org.springframework</groupId>
        <artifactId>spring-context</artifactId>
        <version>5.3.30</version>
    </dependency>

</dependencies>

</project>

```

ApplicationContext.xml:

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository"
          class="com.library.repository.BookRepository"/>

    <bean id="bookService"
          class="com.library.service.BookService">

        <property name="bookRepository"
                  ref="bookRepository"/>

    </bean>

</beans>

```

Main.java

```

package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Main {

    public static void main(String[] args) {

```

```
ApplicationContext ctx =  
    new ClassPathXmlApplicationContext("applicationContext.xml");  
  
BookService service =  
    (BookService) ctx.getBean("bookService");  
  
service.showService();  
}  
}
```

OUTPUT:



The screenshot shows an IDE's Run console. At the top, there's a tab labeled 'Run' and another tab labeled 'Main' with a close button. Below the tabs is a toolbar with icons for running, debugging, and other actions. The main area of the console displays the following output:

```
"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" ...  
Book Service is working.  
Book Repository is working.  
  
Process finished with exit code 0
```

On the left side of the console, there is a vertical toolbar with icons for navigating through the output, including arrows for up and down, a magnifying glass for search, and other standard console navigation icons.